

Negotiating New Permeability Requirements in Bushwick, NY

An SDSS Project for Community Decision Making

Outline

1. Introduction
2. Setting Up The Project
3. SDSS Results
4. Reflections

1. Introduction

Bushwick, NY Keeps Flooding!

In the news:

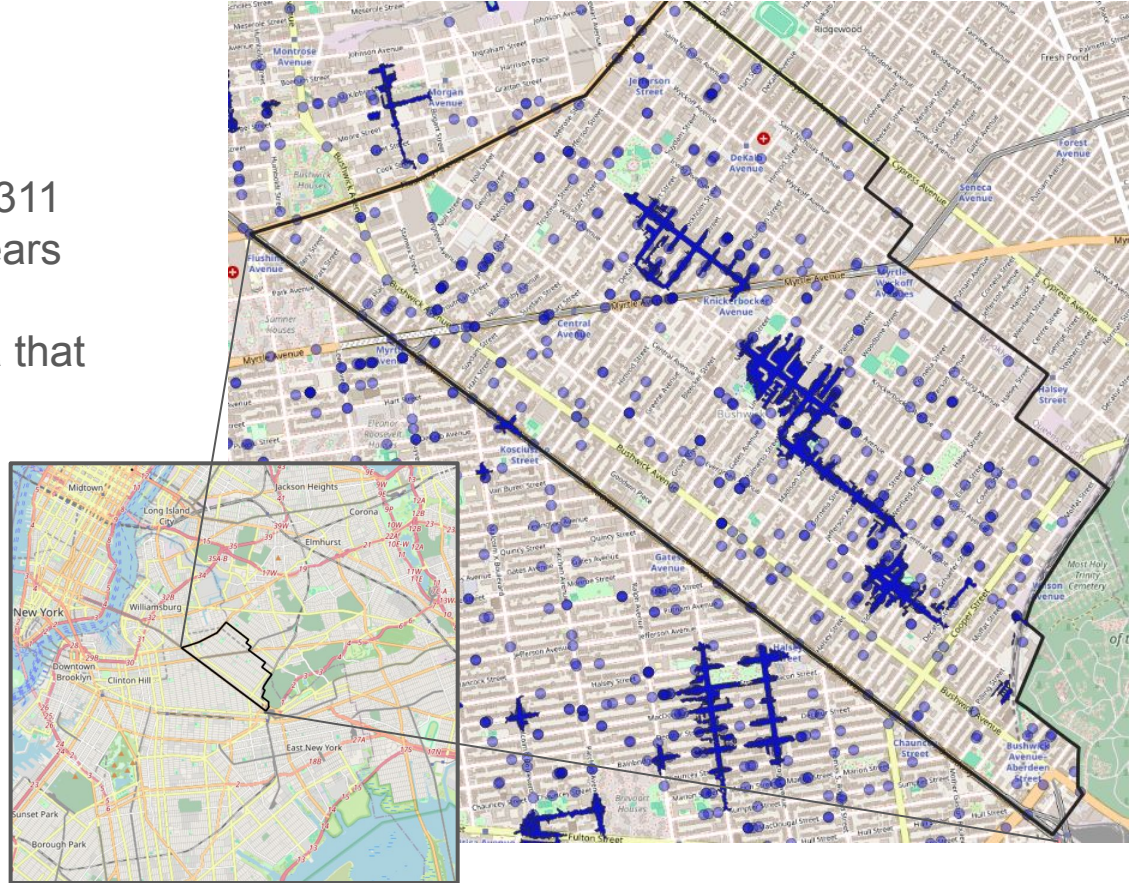
- Low lying area in the center of the neighborhood is prone to flood
- NYC Committed \$390 Million this year to upgrade Bushwick's aging sewer system
- Local Pizza shop owner: "Every time it floods... 'Is today going to be the day that my business loses money?'"



What the Data Shows

Looking at the data:

- Repeated flood-related 311 Calls over the last 15 years
- Flooding projection data that shows Bushwick's problems with flooding will only grow



More Permeability Could Help

Advantages of increased permeability:

- Slows water flow by capturing it in place
- Filters water
- Side benefits: urban heat mitigation, habitat space



What if the City Mandated a 30% Permeability Requirement?

The **Requirement**:

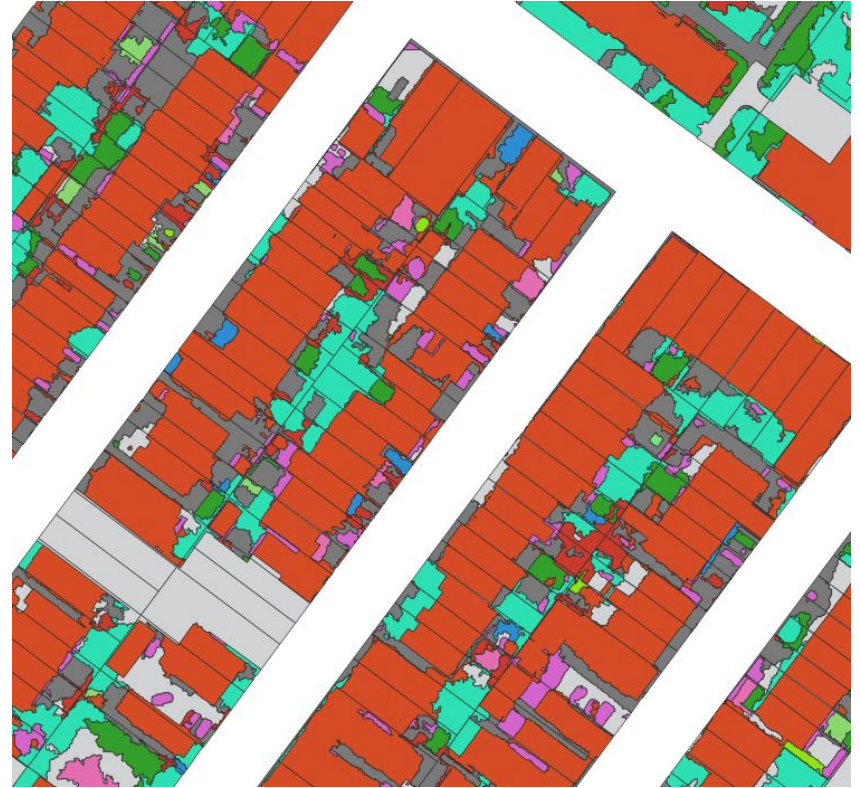
- The neighborhood as a whole must meet 30% permeability
- Each individual property is responsible to meet the 30% requirement

Property Owners' **Options**:

- Convert paved ground areas to green space
- Install green roof space
- Pay into a fund towards increasing green space elsewhere in Bushwick

Current State of Permeability in Bushwick

- **Currently, only 17.3%** of Bushwick is pervious surface
- That means **12.7% more** of Bushwick must become permeable
- **Where will those 6,485,642 sq ft (or 0.23 sq miles) go?**



2. Setting Up The Project

The Stakeholders

- Residents
- Small Business Owners
- Property Owners
- Schools
- The DEP

Stakeholder Concerns | Concern and Metric

1. How much will construction & maintenance **cost**?



1. Estimated **cost to convert** impermeable to permeable: Cheaper to convert is more desirable

2. Will construction **disrupt** my life or business?



2. **Distance to construction**: Building further away is more desirable

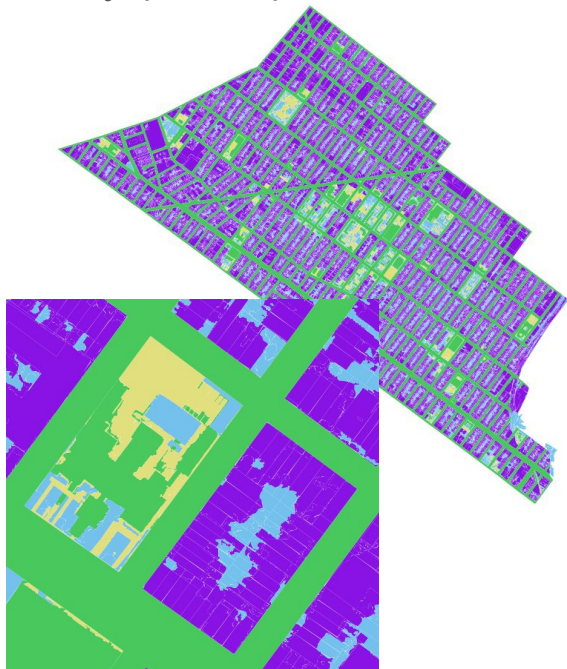
3. Will it be **effective** in preventing flooding?



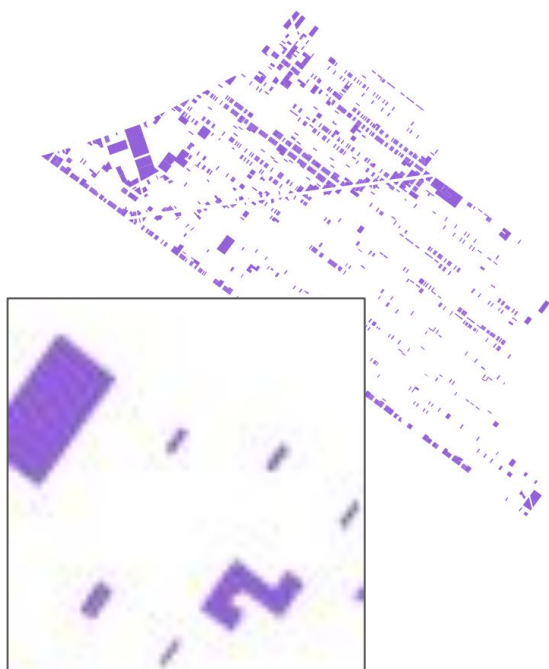
3. **Elevation**: Converting higher (upstream) land is more desirable

Stakeholder Concerns | Data Sources

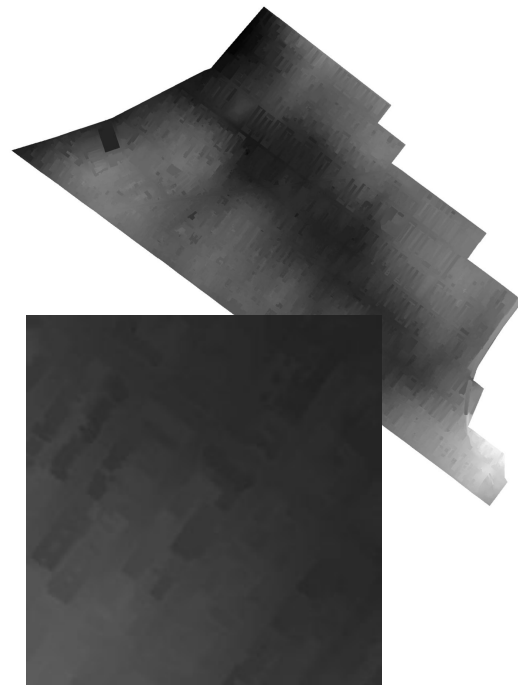
Construction Cost:
DEP Impervious Area
Study (Vector)



NYC PLUTO Map:
Construction Distance



NYC 1ft DEM:
Elevation

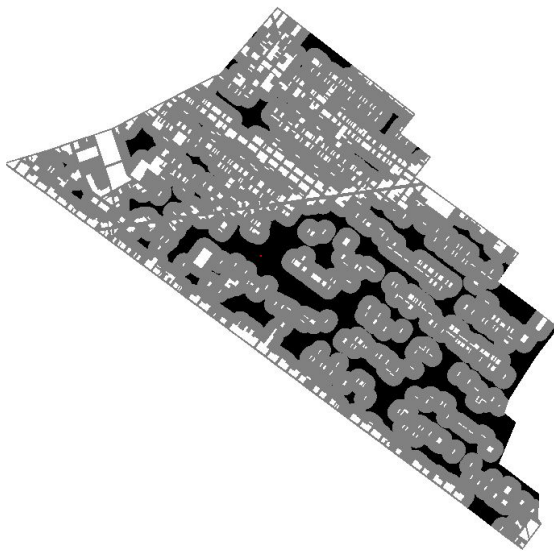


Stakeholder Concerns | 1 to 3 Indexes

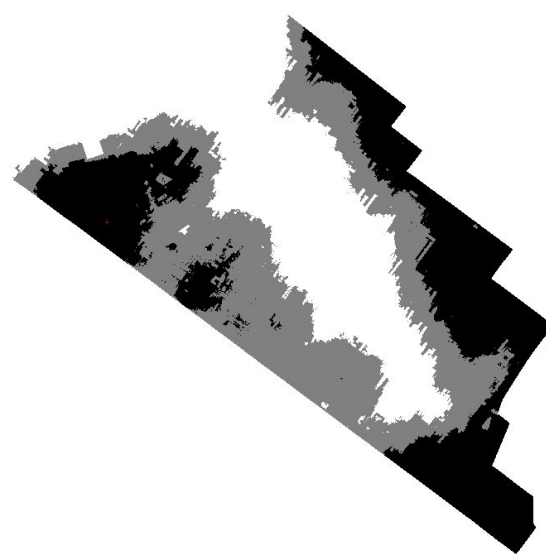
Construction Cost



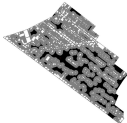
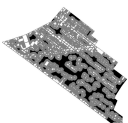
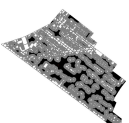
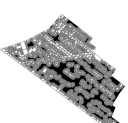
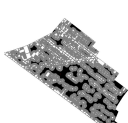
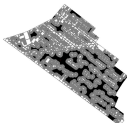
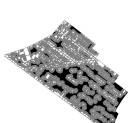
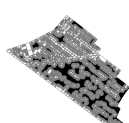
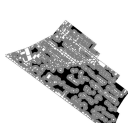
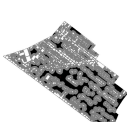
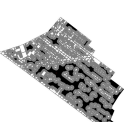
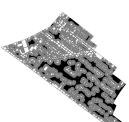
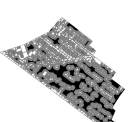
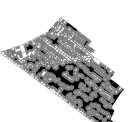
NYC PLUTO Map



NYC 1ft DEM

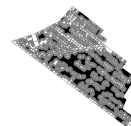


The Weighted Model

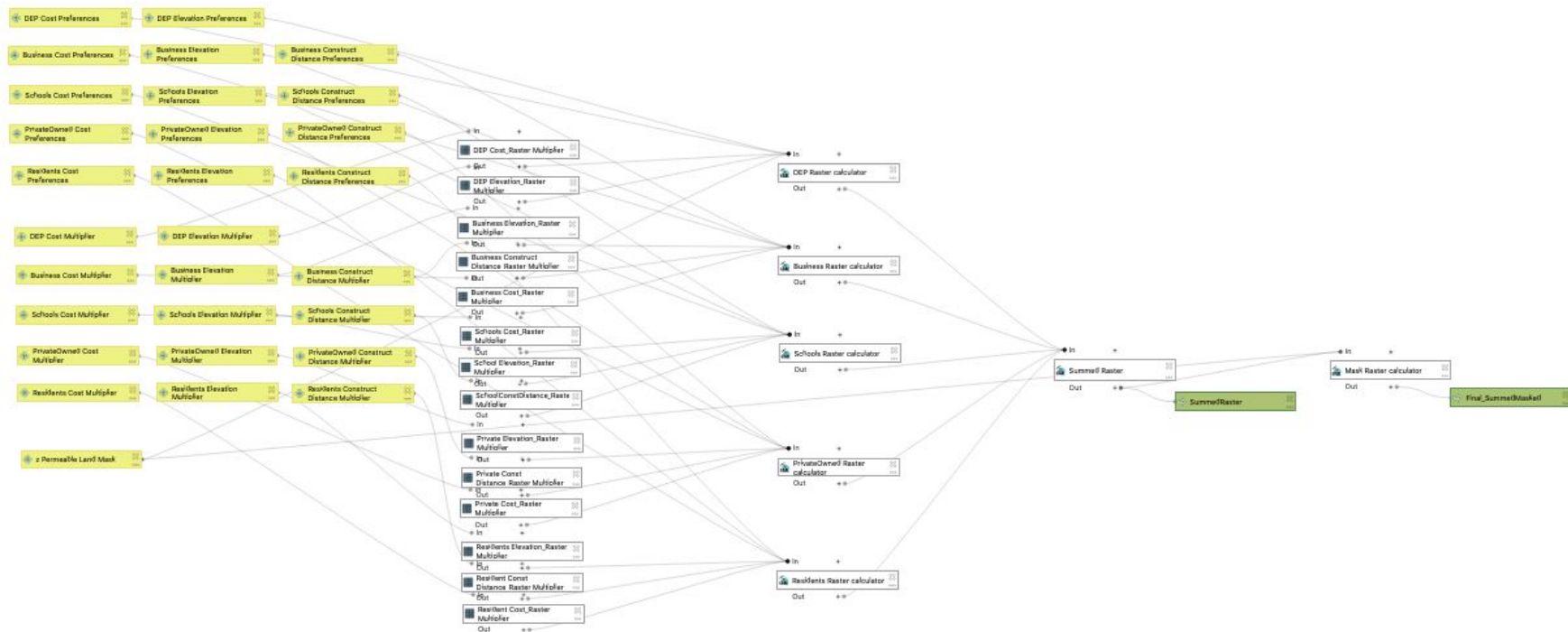
	Residents	Business	Property Owners	Schools	The DEP
Cost	 X 2	 X 2	 X 4	 X 4	 X 3
Distance	 X 4	 X 5	 X 1	 X 3	
Elevation	 X 1	 X 0	 X 2	 X 0	 X 4

X

Permeable
Land Mask

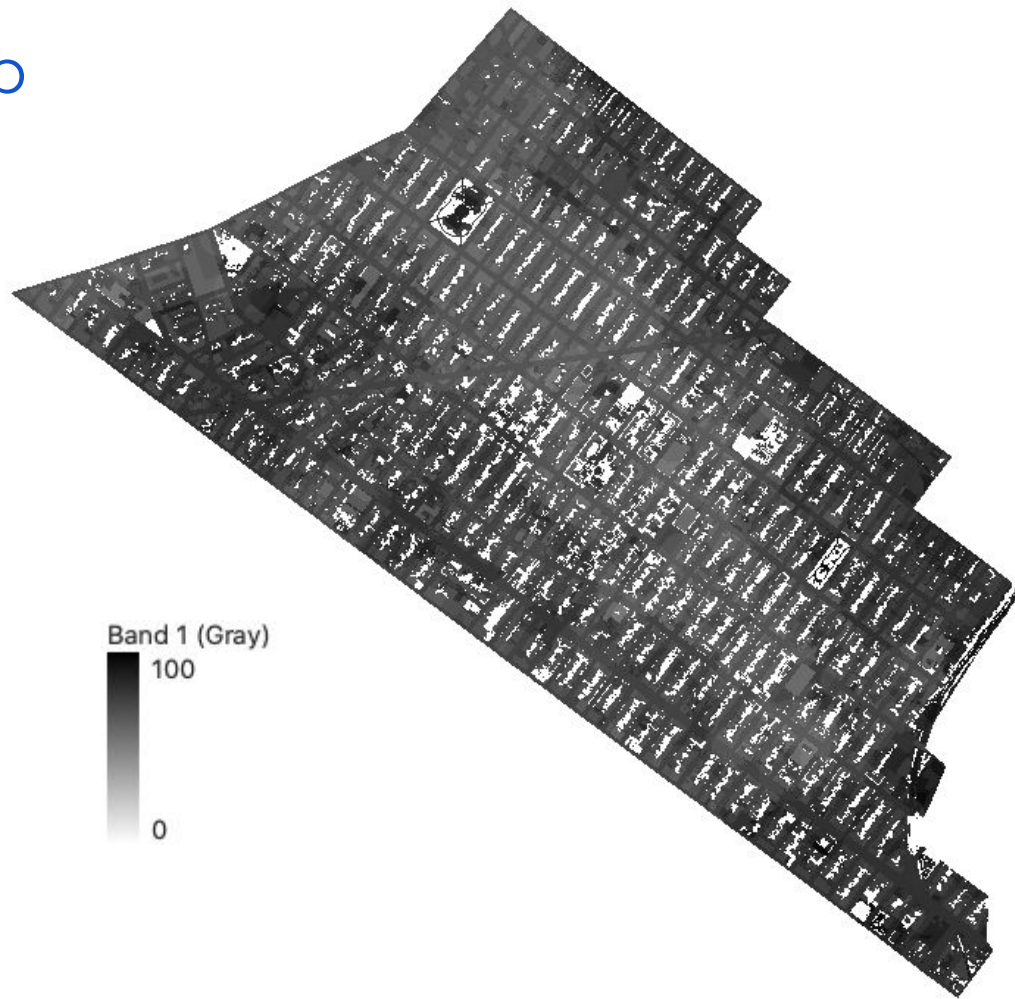


The Weighted Model



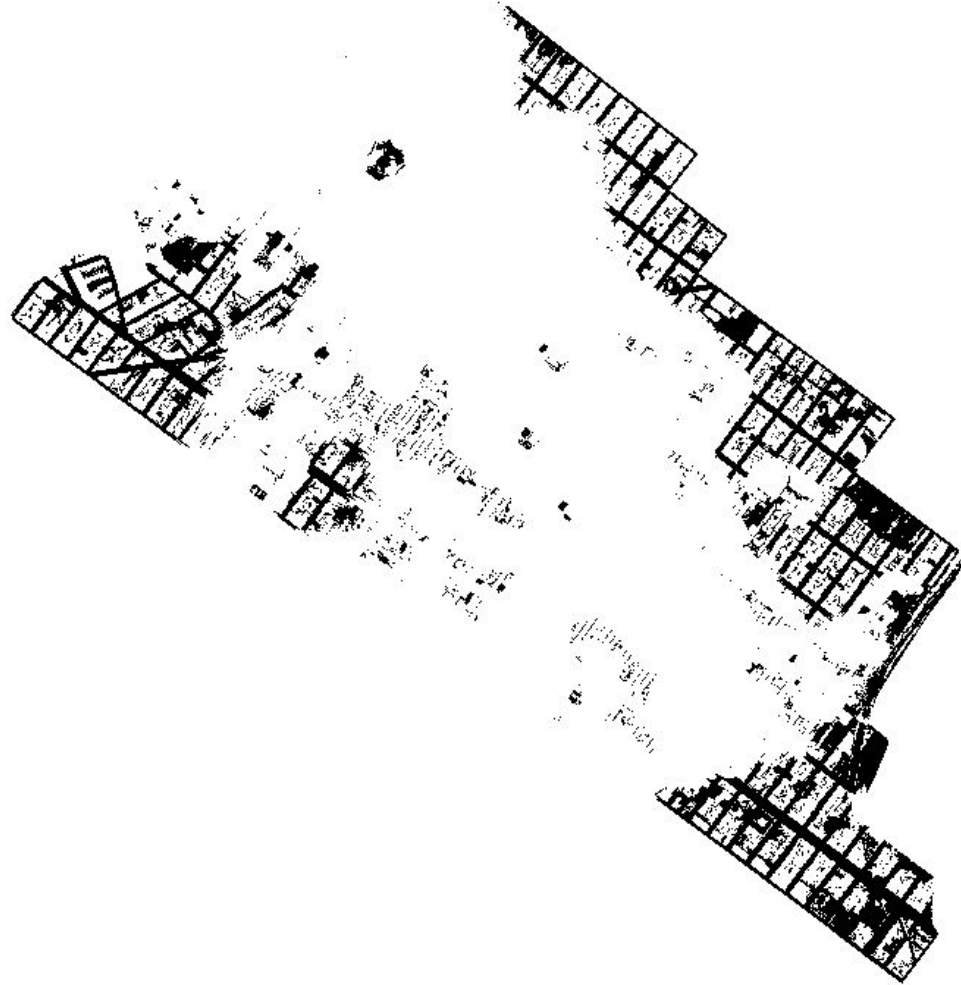
3. SDSS Results

Resulting Map



Resulting Map

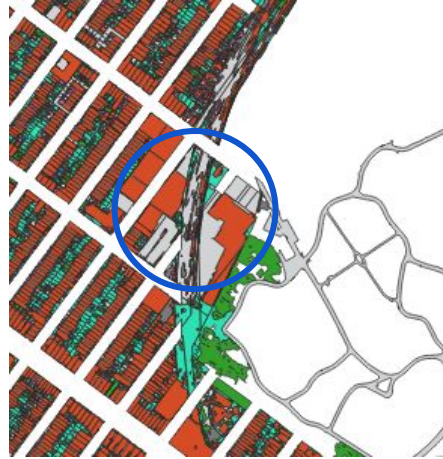
- Places with 80 points or above
- 8,541,805 sq ft total (compared to the 6,485,642 sq ft needed)



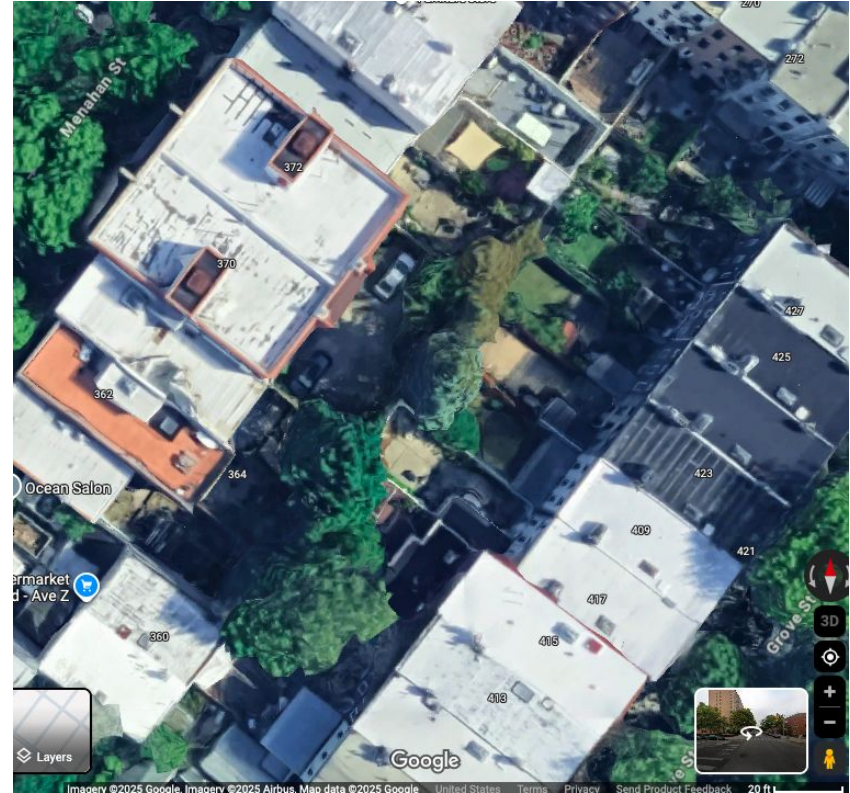
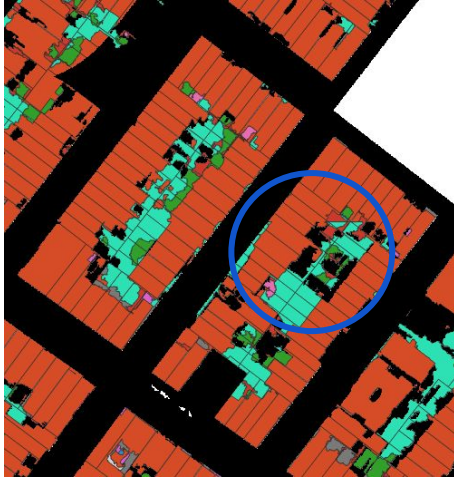
Results | Large Paved Spaces in Higher Areas



Results | Large Paved Spaces in Higher Areas



Results | Residential Spaces In Higher Areas



Results | Streets



4. Reflections

Project Challenges

- Coming up with a topic for an SDSS project
- Simplifying complexity (turning reality into a meaningful model)
- Working with rasters in QGIS (particularly raster calculator)

Further Work

- A “step 2” model that will generate area, cost, elevation stats on the chosen solution
- Refining the metrics
 - A more robust hydrological answer to “effectiveness” of selected areas
 - A more sophisticated analysis of “cost”
- Adding other metrics
 - For example, viability of building at certain sites given the water table level